MICROPROCESSOR AND

MICROCONTROLLER - EC5551

PROJECT REPORT

ELECTRONICS AND COMMUNICATION ENGINEERING

COLLEGE OF ENGINEERING, GUINDY

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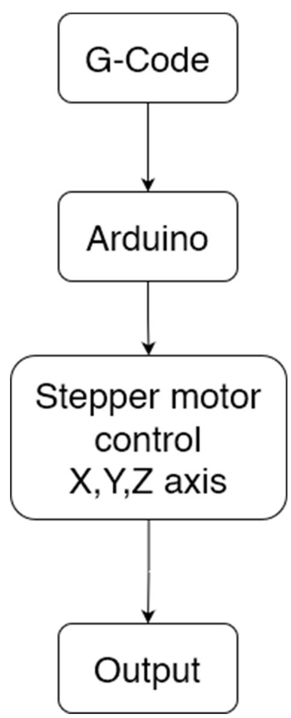
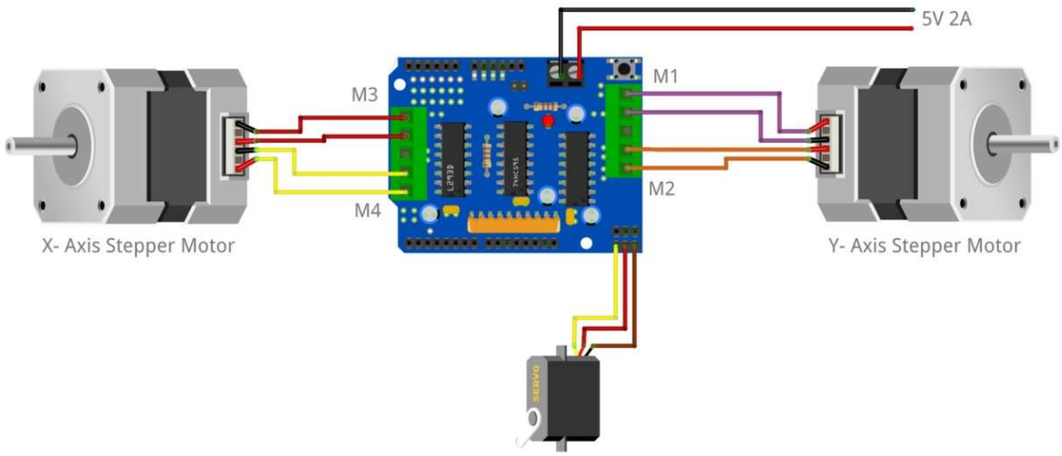
Abstract:

This project involves the design and development of a CNC Plotter, a computer-controlled machine capable of precise and automated 2D drawing, engraving, or plotting tasks. The CNC plotter uses low-cost components such as stepper motors, a microcontroller - Arduino, and a mechanical frame, making it an affordable and accessible alternative to commercial CNC machines. Operating using G-code commands generated through open-source software, provides flexibility for drawing intricate designs and patterns. The project aims to explore the integration of hardware and software to achieve high accuracy and repeatability in motion control. The result is a functional and cost-effective CNC plotter.

Blockdiagram:



Circuit Schematic:



Operation:

Operation of the CNC plotter is as follows:

Power Supply and Initialization:

Supplying power initializes all variables in the code and all ports being used in the arduino in the components: the servo motor, motor driver, two stepper motors.

Uploading the image to be drawn:

A user can upload their GCODE file of choice as the input to be drawn on the paper. The software pronterface was used. This software is connected to the port com7 of the arduino and can control all three axes.

Arduino controls the motors:

In total there are three motors being used, two stepper motors and one servo motor. The stepper motors control the X and Y direction movement, specifically the Y axis is controlled by moving the platform (paper) and the X axis is controlled by moving the pen. Note that these motors are also connected to a servo motor driver.

The Z axis is controlled by a servo motor directly connected to the arduino, this will help in lifting the pen when not needed to be drawn and then placing it at the correct area. A string attached to the motor helps in the process.

Code:

```
#include <Servo.h>
#include <AFMotor.h>

#define LINE_BUFFER_LENGTH 512

char STEP = MICROSTEP;

// Servo position for Up and Down
const int penZUp = 90;
const int penZDown = 50;

// Servo on PWM pin 10
const int penServoPin = 10 ;

// Should be right for DVD steppers, but is not too important here
const int stepsPerRevolution = 48;

// create servo object to control a servo
Servo penServo;

// Initialize steppers for X- and Y-axis using this Arduino pins for the L293D H-bridge
AF_Stepper myStepperY(stepsPerRevolution, 1);
AF_Stepper myStepperX(stepsPerRevolution, 2);

/* Structures, global variables         */
struct point {
  float x;
  float y;
  float z;
```

```
};

// Current position of plothead
struct point actuatorPos;

//  Drawing settings, should be OK
float StepInc = 1;
int StepDelay = 0;
int LineDelay = 0;
int penDelay = 50;

// Motor steps to go 1 millimeter.
// Use test sketch to go 100 steps. Measure the length of line.
// Calculate steps per mm. Enter here.
float StepsPerMillimeterX = 100.0;
float StepsPerMillimeterY = 100.0;

// Drawing robot limits, in mm
// OK to start with. Could go up to 50 mm if calibrated well.
float Xmin = 0;
float Xmax = 40;
float Ymin = 0;
float Ymax = 40;
float Zmin = 0;
float Zmax = 1;

float Xpos = Xmin;
float Ypos = Ymin;
float Zpos = Zmax;

// Set to true to get debug output.
boolean verbose = false;

//  Needs to interpret
```

```
//  G1 for moving
//  G4 P300 (wait 150ms)
//  M300 S30 (pen down)
//  M300 S50 (pen up)
//  Discard anything with a (
//  Discard any other command!

/**********************
   void setup() - Initialisations
 ***********************/
void setup() {
  //  Setup

  Serial.begin( 9600 );

  penServo.attach(penServoPin);
  penServo.write(penZUp);
  delay(100);

  // Decrease if necessary
  myStepperX.setSpeed(600);

  myStepperY.setSpeed(600);

  //  Set & move to initial default position
  // TBD

  //  Notifications!!!
  Serial.println("Mini CNC Plotter alive and kicking!");
  Serial.print("X range is from ");
  Serial.print(Xmin);
  Serial.print(" to ");
  Serial.print(Xmax);
  Serial.println(" mm.");
```

```
  Serial.print("Y range is from ");
  Serial.print(Ymin);
  Serial.print(" to ");
  Serial.print(Ymax);
  Serial.println(" mm.");
}

/**********************
   void loop() - Main loop
 ***********************/
void loop()
{

  delay(100);
  char line[ LINE_BUFFER_LENGTH ];
  char c;
  int lineIndex;
  bool lineIsComment, lineSemiColon;

  lineIndex = 0;
  lineSemiColon = false;
  lineIsComment = false;

  while (1) {

        // Serial reception - Mostly from Grbl, added semicolon support
        while ( Serial.available() > 0 ) {
        c = Serial.read();
        if (( c == '\n') || (c == '\r') ) {        // End of line reached
        if ( lineIndex > 0 ) {                     // Line is complete. Then execute!
        line[ lineIndex ] = '\0';                  // Terminate string
        if (verbose) {
        Serial.print( "Received : ");
        Serial.println( line );
```

```
}
processIncomingLine( line, lineIndex );
lineIndex = 0;
}
else {
// Empty or comment line. Skip block.
}
lineIsComment = false;
lineSemiColon = false;
Serial.println("ok");
}
else {
if ( (lineIsComment) || (lineSemiColon) ) {   // Throw away all comment characters
if ( c == ')' )  lineIsComment = false;           // End of comment. Resume line.
}
else {
if ( c <= ' ' ) {                                // Throw away whitepace and control characters
}
else if ( c == '/' ) {                           // Block delete not supported. Ignore character.
}
else if ( c == '(' ) {                           // Enable comments flag and ignore all characters until ')' or EOL.
lineIsComment = true;
}
else if ( c == ';' ) {
lineSemiColon = true;
}
else if ( lineIndex >= LINE_BUFFER_LENGTH - 1 ) {
Serial.println( "ERROR - lineBuffer overflow" );
lineIsComment = false;
lineSemiColon = false;
}
else if ( c >= 'a' && c <= 'z' ) {      // Upcase lowercase
line[ lineIndex++ ] = c - 'a' + 'A';
}
```

```
            else {
            line[ lineIndex++ ] = c;
            }
            }
            }
            }
  }
}

void processIncomingLine( char* line, int charNB ) {
  int currentIndex = 0;
  char buffer[ 64 ];                              // Hope that 64 is enough for 1 parameter
  struct point newPos;

  newPos.x = 0.0;
  newPos.y = 0.0;

  //  Needs to interpret
  //  G1 for moving
  //  G4 P300 (wait 150ms)
  //  G1 X60 Y30
  //  G1 X30 Y50
  //  M300 S30 (pen down)
  //  M300 S50 (pen up)
  //  Discard anything with a (
  //  Discard any other command!

  while ( currentIndex < charNB ) {
            switch ( line[ currentIndex++ ] ) {              // Select command, if any
            case 'U':
            penUp();
            break;
            case 'D':
            penDown();
```

```
break;
case 'G':
buffer[0] = line[ currentIndex++ ];          // /!\ Dirty - Only works with 2 digit
commands
//         buffer[1] = line[ currentIndex++ ];
//         buffer[2] = '\0';
buffer[1] = '\0';

switch ( atoi( buffer ) ) {          // Select G command
case 0:                              // G00 & G01 - Movement or fast movement. Same here
case 1:
// /!\ Dirty - Suppose that X is before Y
char* indexX = strchr( line + currentIndex, 'X' ); // Get X/Y position in the string (if any)
char* indexY = strchr( line + currentIndex, 'Y' );
if ( indexY <= 0 ) {
newPos.x = atof( indexX + 1);
newPos.y = actuatorPos.y;
}
else if ( indexX <= 0 ) {
newPos.y = atof( indexY + 1);
newPos.x = actuatorPos.x;
}
else {
newPos.y = atof( indexY + 1);
indexY = '\0';
newPos.x = atof( indexX + 1);
}
drawLine(newPos.x, newPos.y );
//         Serial.println("ok");
actuatorPos.x = newPos.x;
actuatorPos.y = newPos.y;
break;
}
break;
```

```
case 'M':
buffer[0] = line[ currentIndex++ ];              // /!\ Dirty - Only works with 3 digit commands
buffer[1] = line[ currentIndex++ ];
buffer[2] = line[ currentIndex++ ];
buffer[3] = '\0';
switch ( atoi( buffer ) ) {
case 300:
{
char* indexS = strchr( line + currentIndex, 'S' );
float Spos = atof( indexS + 1);
//            Serial.println("ok");
if (Spos == 30) {
penDown();
}
if (Spos == 50) {
penUp();
}
break;
}
case 114:                           // M114 - Repport position
Serial.print( "Absolute position : X = " );
Serial.print( actuatorPos.x );
Serial.print( "  -  Y = " );
Serial.println( actuatorPos.y );
break;
default:
Serial.print( "Command not recognized : M");
Serial.println( buffer );
}
}
  }
}

void drawLine(float x1, float y1) {
```

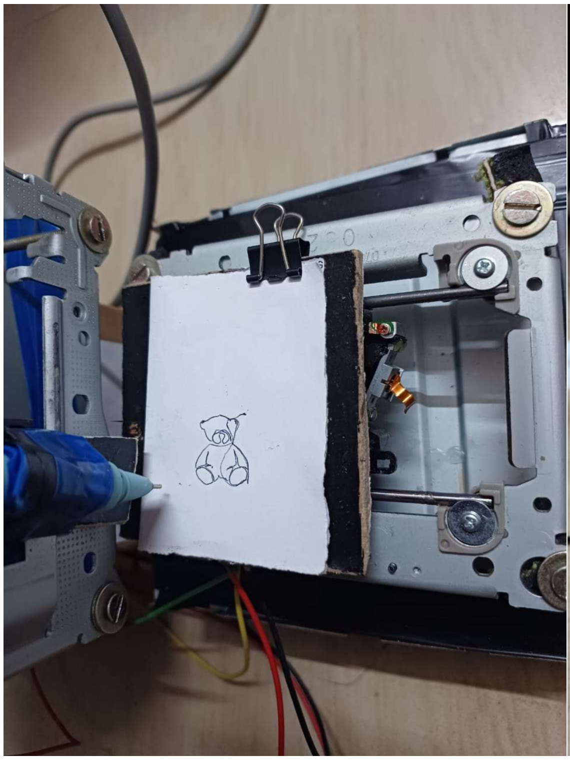
```
if (verbose)
{
	Serial.print("fx1, fy1: ");
	Serial.print(x1);
	Serial.print(",");
	Serial.print(y1);
	Serial.println("");
}

//  Bring instructions within limits
if (x1 >= Xmax) {
	x1 = Xmax;
}
if (x1 <= Xmin) {
	x1 = Xmin;
}
if (y1 >= Ymax) {
	y1 = Ymax;
}
if (y1 <= Ymin) {
	y1 = Ymin;
}

if (verbose)
{
	Serial.print("Xpos, Ypos: ");
	Serial.print(Xpos);
	Serial.print(",");
	Serial.print(Ypos);
	Serial.println("");
}

if (verbose)
{
```

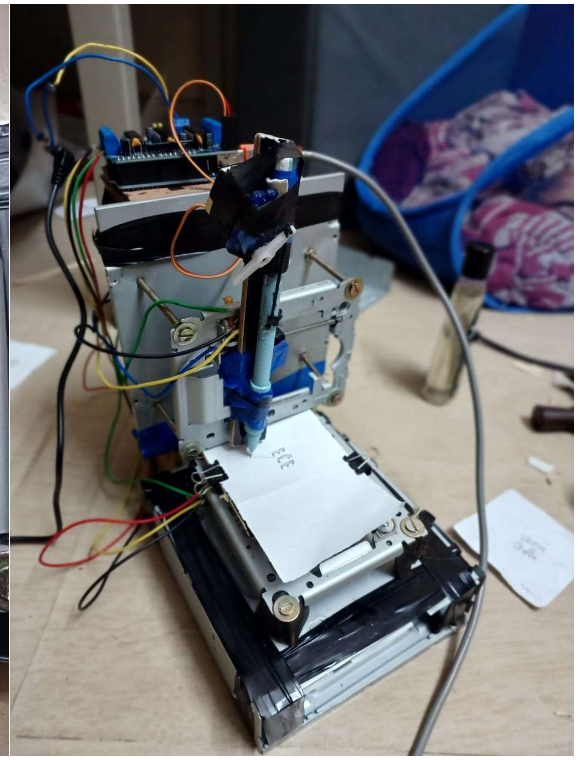
```
        Serial.print("x1, y1: ");
        Serial.print(x1);
        Serial.print(",");
        Serial.print(y1);
        Serial.println("");
}

//  Convert coordinates to steps
x1 = (int)(x1 * StepsPerMillimeterX);
y1 = (int)(y1 * StepsPerMillimeterY);
float x0 = Xpos;
float y0 = Ypos;

//  Let's find out the change for the coordinates
long dx = abs(x1 - x0);
long dy = abs(y1 - y0);
int sx = x0 < x1 ? StepInc : -StepInc;
int sy = y0 < y1 ? StepInc : -StepInc;

long i;
long over = 0;

if (dx > dy) {
        for (i = 0; i < dx; ++i) {
        myStepperX.onestep(sx, STEP);
        over += dy;
        if (over >= dx) {
        over -= dx;
        myStepperY.onestep(sy, STEP);
        }
        delay(StepDelay);
        }
}
else {
```

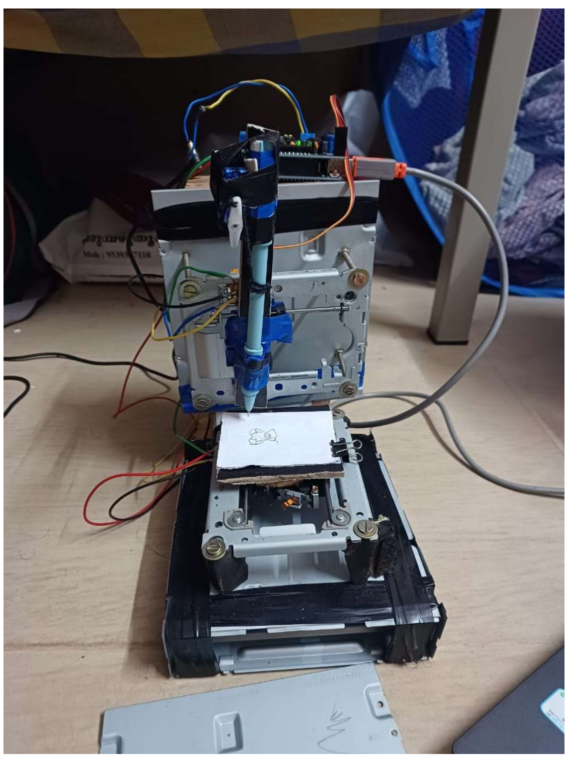
```
        for (i = 0; i < dy; ++i) {
        myStepperY.onestep(sy, STEP);
        over += dx;
        if (over >= dy) {
        over -= dy;
        myStepperX.onestep(sx, STEP);
        }
        delay(StepDelay);
        }
}

if (verbose)
{
        Serial.print("dx, dy:");
        Serial.print(dx);
        Serial.print(",");
        Serial.print(dy);
        Serial.println("");
}

if (verbose)
{
        Serial.print("Going to (");
        Serial.print(x0);
        Serial.print(",");
        Serial.print(y0);
        Serial.println(")");
}

//  Delay before any next lines are submitted
delay(LineDelay);
//  Update the positions
Xpos = x1;
Ypos = y1;
```

```
}

// Raises pen
void penUp() {
  penServo.write(penZUp);
  delay(penDelay);
  Zpos = Zmax;
  digitalWrite(15, LOW);
  digitalWrite(16, HIGH);
  if (verbose) {
        Serial.println("Pen up!");
  }
}
// Lowers pen
void penDown() {
  penServo.write(penZDown);
  delay(penDelay);
  Zpos = Zmin;
  digitalWrite(15, HIGH);
  digitalWrite(16, LOW);
  if (verbose) {
        Serial.println("Pen down.");
  }
}
```

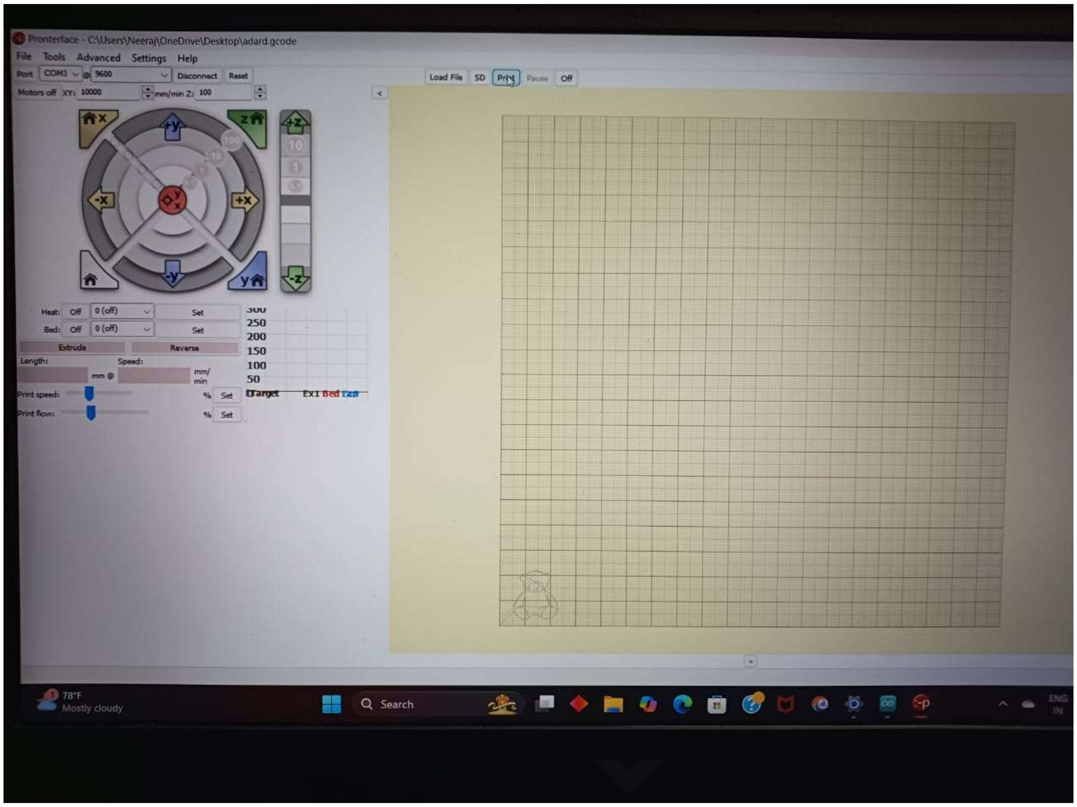
CNC PLOTTER :







PRONTER FACE SOFTWARE :



Result:

The CNC Plotter project successfully demonstrated a functional, low-cost, and precise 2D plotting machine. The system accurately translated G-code commands into mechanical motion, enabling the plotting of intricate designs and patterns with high repeatability. The project effectively integrated hardware and software components, showcasing key principles of automation.